Phase 2 (part 3): (Non)Linear Curve-Fitting

1.] A lot of real-world data is exponential in nature and so fitting a function of the form $f(x) = ae^{bx}$ is necessary. Show that for any set of data, fitting this curve can be transformed into a linear fitting problem.

Let $\vec{x} = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{n-1} \end{bmatrix}$ and $\vec{y} = \begin{bmatrix} y_0 \\ y_1 \\ \vdots \\ y_{n-1} \end{bmatrix}$ represent a set of n data points. Then

$f(x_i) = ae^{bx_i} = y_i$ for all i. But when can write

$$ae^{bx_i} = y_i \Rightarrow \ln(a) + bx_i = \ln(y_i) \text{ for } i = 0, 1, 2, \ldots, n-1.$$

In vector form:

$$\begin{cases} \ln(a) + bx_0 = \ln(y_0) \\ \ln(a) + bx_1 = \ln(y_1) \\ \vdots \\ \ln(a) + bx_{n-1} = \ln(y_{n-1}) \end{cases} \Rightarrow \ln(a)\begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} + b\begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{n-1} \end{bmatrix} = \begin{bmatrix} \ln(y_0) \\ \ln(y_1) \\ \vdots \\ \ln(y_{n-1}) \end{bmatrix} \Rightarrow \underbrace{\begin{bmatrix} 1 & x_0 \\ 1 & x_1 \\ \vdots & \vdots \\ 1 & x_{n-1} \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} \ln(a) \\ b \end{bmatrix}}_{\vec{c}} = \underbrace{\begin{bmatrix} \ln(y_0) \\ \ln(y_1) \\ \vdots \\ \ln(y_{n-1}) \end{bmatrix}}_{\vec{y}}$$

2.] A lot of real-world data is logistic in nature and so fitting a function of the form $f(x) = \dfrac{1}{1 + ae^{-bx}}$ is necessary. Here, a and b are both strictly positive. Show that for any set of data, fitting this curve can be transformed into a linear fitting problem.

For each (x_i, y_i), we have

$$f(x_i) = \frac{1}{1 + ae^{-bx_i}} = y_i$$

$$\Rightarrow 1 + ae^{-bx_i} = \frac{1}{y_i}$$

$$\Rightarrow ae^{-bx_i} = \frac{1}{y_i} - 1$$

$$\Rightarrow \ln(a) - bx_i = \ln\left(\frac{1}{y_i} - 1\right)$$

$$\Rightarrow \ln(a)\begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} + b\begin{bmatrix} -x_0 \\ -x_1 \\ \vdots \\ -x_{n-1} \end{bmatrix} = \begin{bmatrix} \ln(1/y_0 - 1) \\ \ln(1/y_1 - 1) \\ \vdots \\ \ln(1/y_{n-1} - 1) \end{bmatrix}$$

$$\Rightarrow \underbrace{\begin{bmatrix} 1 & -x_0 \\ 1 & -x_1 \\ \vdots & \vdots \\ 1 & -x_{n-1} \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} \ln(a) \\ b \end{bmatrix}}_{\vec{c}} = \underbrace{\begin{bmatrix} \ln\left(\frac{1}{y_0} - 1\right) \\ \ln\left(\frac{1}{y_1} - 1\right) \\ \vdots \\ \ln\left(\frac{1}{y_{n-1}} - 1\right) \end{bmatrix}}_{\vec{y}}$$

3.] Similar to the previous example, we may want to fit a "Witch of Agnesi" function to specific data. This function has the form $f(x) = \dfrac{a}{bx^2+1}$, and looks similar to a Normal curve. Here, $a > 0$ is a parameter that controls the height of the centered peak, and $b > 0$ controls the width. Show that a function of this form can be fit to data by transforming it into a linear problem.

For each (x_i, y_i), we have

$$f(x_i) = \frac{a}{bx_i^2+1} = y_i$$

$$\Rightarrow \quad \frac{bx_i^2+1}{a} = \frac{1}{y_i}$$

$$\Rightarrow \quad \frac{b}{a}x_i^2 + \frac{1}{a} = \frac{1}{y_i}$$

$$\Rightarrow \quad \frac{1}{a} + \frac{b}{a}x_i^2 = \frac{1}{y_i}$$

$$\frac{1}{a}\begin{bmatrix}1\\1\\\vdots\\1\end{bmatrix} + \frac{b}{a}\begin{bmatrix}x_0^2\\x_1^2\\\vdots\\x_{n-1}^2\end{bmatrix} = \begin{bmatrix}1/y_0\\1/y_1\\\vdots\\1/y_{n-1}\end{bmatrix}$$

$$\Rightarrow \underbrace{\begin{bmatrix}1 & x_0^2\\1 & x_1^2\\\vdots & \vdots\\1 & x_{n-1}^2\end{bmatrix}}_{A}\underbrace{\begin{bmatrix}1/a\\b/a\end{bmatrix}}_{\vec{c}} = \underbrace{\begin{bmatrix}1/y_0\\1/y_1\\\vdots\\1/y_{n-1}\end{bmatrix}}_{\vec{y}}$$

4.] Data may following a periodic or sinusoidal pattern. Therefore, fitting sinusoidal data to a function such as

$$f(x) = c_0 + c_1\sin(x) + c_2\cos(x) + c_3\sin(2x) + c_4\cos(2x),$$

may be beneficial. Show that fitting this function to data is also a linear least-squares problem.

For each (x_i, y_i), we have

$$f(x_i) = c_0 + c_1\sin(x_i) + c_2\cos(x_i) + c_3\sin(2x_i) + c_4\cos(2x_i) = y_i$$

$$\Rightarrow c_0\begin{bmatrix}1\\1\\\vdots\\1\end{bmatrix} + c_1\begin{bmatrix}\sin(x_0)\\\sin(x_1)\\\vdots\\\sin(x_{n-1})\end{bmatrix} + c_2\begin{bmatrix}\cos(x_0)\\\cos(x_1)\\\vdots\\\cos(x_{n-1})\end{bmatrix} + c_3\begin{bmatrix}\sin(2x_0)\\\sin(2x_1)\\\vdots\\\sin(2x_{n-1})\end{bmatrix} + c_4\begin{bmatrix}\cos(2x_0)\\\cos(2x_1)\\\vdots\\\cos(2x_{n-1})\end{bmatrix} = \begin{bmatrix}y_0\\y_1\\\vdots\\y_{n-1}\end{bmatrix}$$

$$\Rightarrow \underbrace{\begin{bmatrix}1 & \sin(x_0) & \sin(x_0) & \sin(2x_0) & \cos(2x_0)\\1 & \sin(x_1) & \sin(x_1) & \sin(2x_1) & \cos(2x_1)\\\vdots & \vdots & \vdots & \vdots & \vdots\\1 & \sin(x_{n-1}) & \sin(x_{n-1}) & \sin(2x_{n-1}) & \cos(2x_{n-1})\end{bmatrix}}_{A}\underbrace{\begin{bmatrix}c_0\\c_1\\c_2\\c_3\\c_4\end{bmatrix}}_{\vec{c}} = \underbrace{\begin{bmatrix}y_0\\y_1\\\vdots\\y_{n-1}\end{bmatrix}}_{\vec{y}}$$